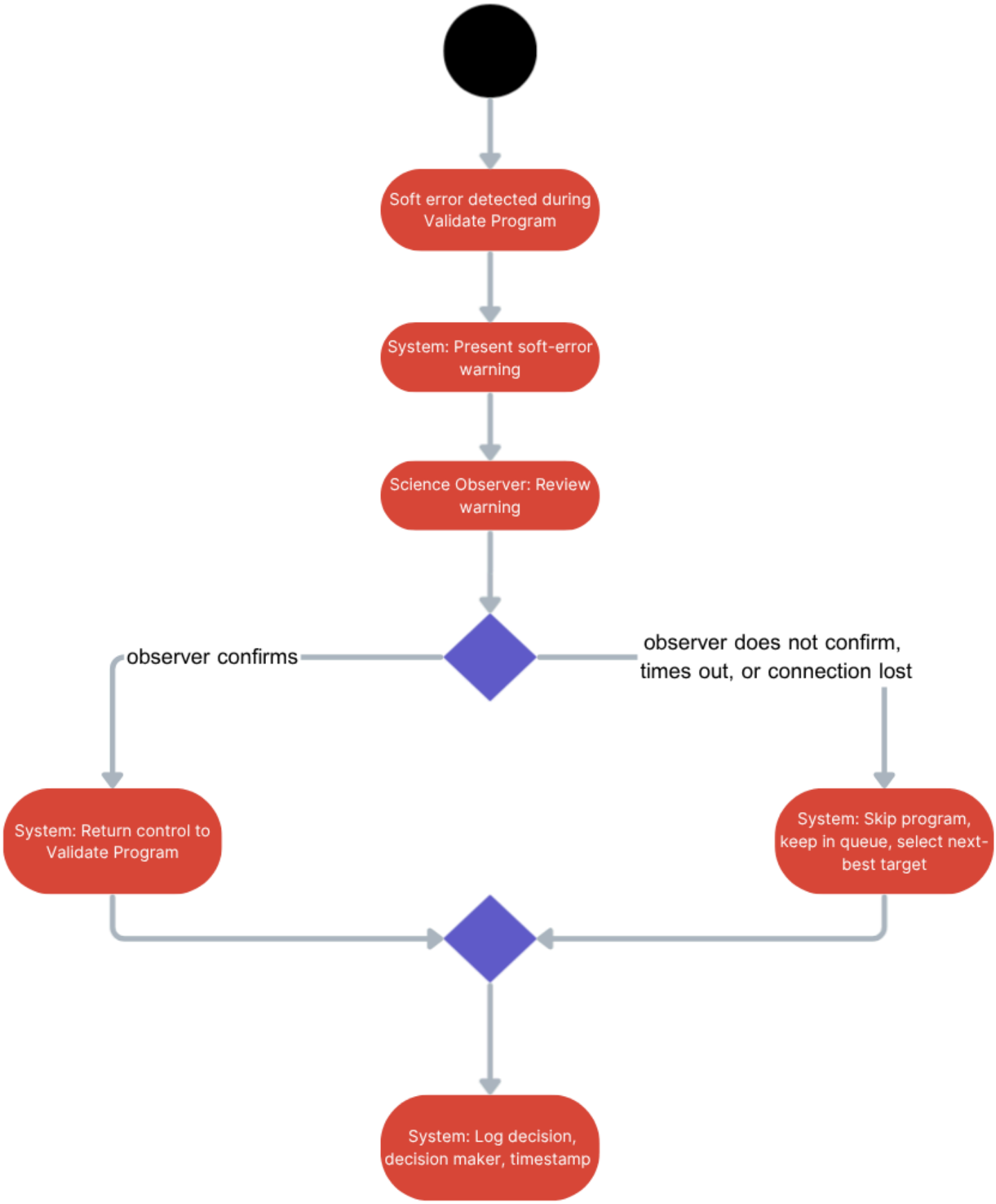
1. Confirm Soft-Error Warning

Use Case ID:	UC-01		
Use Case Name:	Confirm Soft-Error Warning		
Process Owner:	Gemini Science Operations	Last Updated By:	Woraneti
Date Created:	9/15/2026	Date Last Updated:	9/15/2026
Business Actor:	Science Observer		
Description	During program validation the system distinguishes between hard physical errors and soft errors. Hard errors — such as a target below the horizon or a filter not mounted in the wheel — are rejected automatically and never reach this use case. Soft errors — such as observing at a high airmass where image quality degrades — are not rejected: the system raises a warning and the Science Observer decides whether the observation proceeds as planned or stops.		
Trigger	Trigger: Program validation detects one or more soft errors in an observing program that is being prepared for execution. Type: External		
Relationships	Association: Science Observer Include: None Extend: Validate Program (this use case extends Validate Program) Generalization: None		
Preconditions:	1. The Science Observer is logged in and holds a role authorised to act on the program. 2. The system is at the Observing operational level. 3. Validate Program is executing for an observing program that has been submitted for execution. 4. Validation has detected at least one soft error. 5. Validation has detected no hard physical errors — a program containing a hard error is rejected before reaching this use case.		
Postconditions:	1. The decision, the identity of the decision maker, and an exact timestamp are recorded and retained permanently. 2. If the observer proceeds: control returns to Validate Program, which completes and allows the program to continue toward		

	execution. 3. If the observer does not proceed: the program does not continue to execute; it is skipped, returned to the pending list, and the system selects the next-best target.
Performance Goal:	The observer's confirmation is a command to the system and must be accepted or rejected within 2 seconds, before the resulting action occurs.
Basic Workflow:	1. The Validate Program reaches the extension point at which a soft error has been detected. 2. The system presents the soft-error warning to the Science Observer, identifying the program and the condition that raised it. 3. The Science Observer reviews the warning against the scientific goals of the program. 4. The Science Observer confirms that the observation should proceed. 5. The system records the decision, the decision maker, and the timestamp. 6. The system returns control to Validate Program, which completes validation.
Alternative Workflow:	2a. Validation detects more than one soft error in the same program. 4a. The Science Observer decides not to proceed. 4b. The Science Observer's connection is lost before a decision is made.
Category:	Observation Management
Risks:	1. The observer approves a warning under time pressure (the directorate's usable-night KPI) without fully weighing the scientific trade-off, leading to degraded data that is only discovered during analysis. 2. Warning fatigue — if soft-error warnings recur often for common conditions (e.g. routine high-airmass targets), observers may confirm reflexively without reading the specific condition each time. 3. The system's timeout for "too long to decide" is not specified by the PO; if the team sets it too short, valid decisions get cut off; if too long, it

	conflicts with the directorate's stated priority of minimizing idle time. 4. If the team's proposal is "confirm all warnings at once" for a program with multiple soft errors, an observer may miss one specific issue buried among several.
Possibilities:	1. Let observers set default responses for well-understood, recurring warning types (e.g. "always proceed on airmass warnings above a given threshold for this program"), subject to periodic review. 2. Notify the remote Astronomer of the decision after the fact, even though they cannot act on it directly — keeps the program owner informed. 3. Track each observer's warning-decision history to support the audit and training use cases already in D1 § Logging. 4. A severity-scaled timeout (a marginal soft warning gets less waiting time than one closer to the hard-error boundary).


Soft error detected during Validate Program
System: Present soft-error warning
Science Observer: Review warning
observer confirms
observer does not confirm, times out, or connection lost
System: Return control to Validate Program
System: Skip program, keep in queue, select next-best target
System: Log decision, decision maker, timestamp

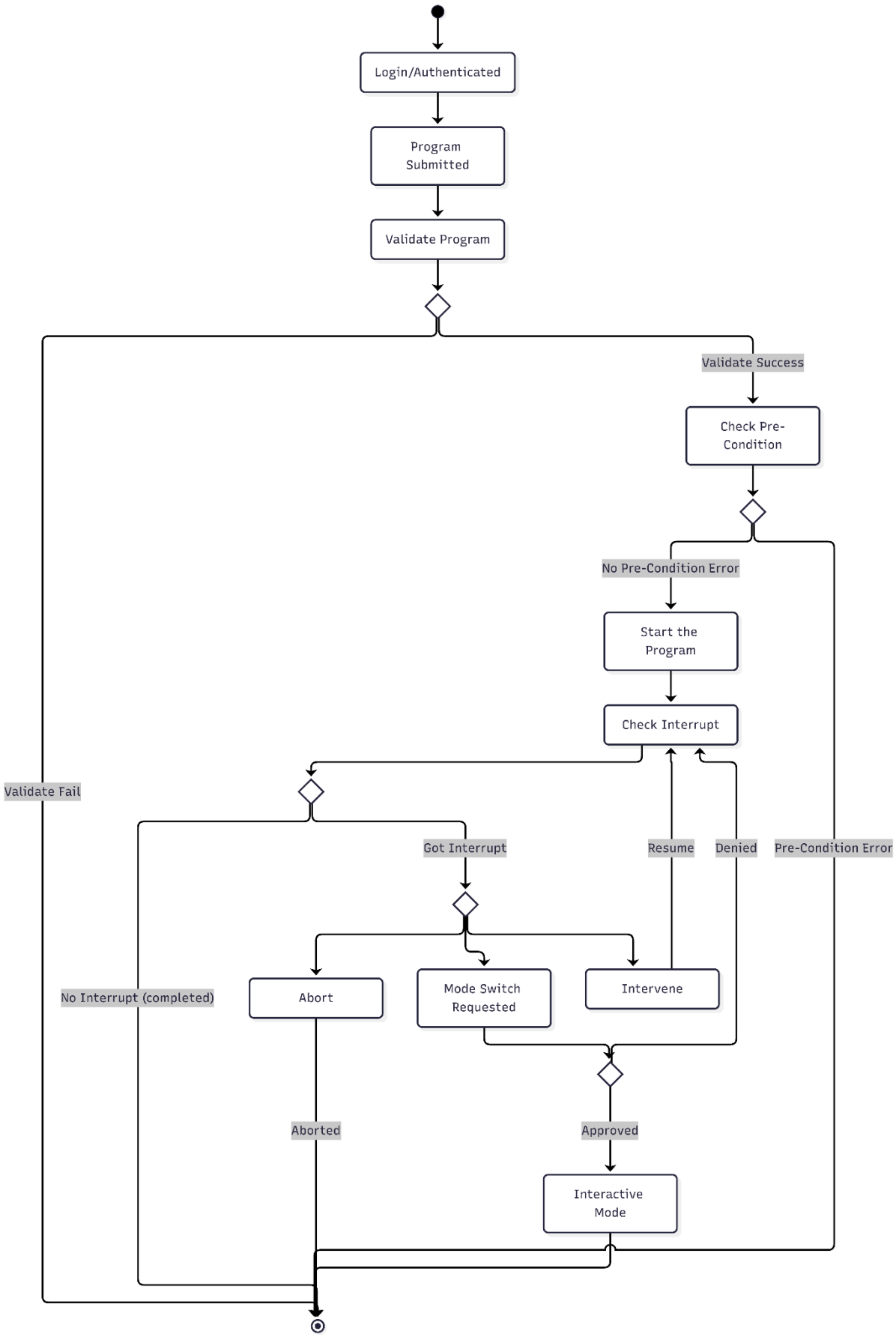
3. Start Observing Program

Use Case ID:	UC-03		
Use Case Name:	Start Observing Program		
Process Owner:	Gemini Science Operations	Last Updated By:	Phoonyaphat Junsiri, Barameeporn Niyomtrong
Date Created:	15/09/2026	Date Last Updated:	15/09/2026
Business Actor:	Telescope Operator		
Description:	This use case describes how the Telescope Operator starts and controls execution of an observing program, and how a Science Observer may request to switch to Interactive Mode during execution to issue commands directly.		
Trigger	The telescope operator commands the system to begin observing. Type : External		
Relationship	Association : Telescope Operator, Science Observer Include : Lock Instrument Configuration, Track Allocated Time, Validate Program, Check Instrument Configuration, Execute Command Sequence, Retrieve Guide and Standard Stars, Store Raw Data Extend : Run Required Calibration, Apply Fault Recovery Procedure, Intervene in Running Observation, Abort Exposure Generalization : None		
Preconditions:	1. Telescope operators must be authenticated/logged in first. 2. An observing program must be submitted. 3. The instrument to be used must already be configured. 4. The Weather Monitoring System must be in a state that allows observing to start. 5. No other subsystem may be locked/in control (or: "No other subsystem should be holding a lock or control").		
Postconditions:	1. The observing program is actively executing under Telescope Operator control (in Planned, Queue, or Service mode).		

	2. If a mode switch was requested and approved, control has been handed to the Science Observer in Interactive Mode. 3. The system has logged the start of execution (for audit/tracking purposes). 4. Allocated observing time tracking has been updated to reflect the session in progress.
Performance Goal:	1. The system shall begin executing a validated observing program within 2 seconds of the Telescope Operator's start command. 2. Mode switch to Interactive Mode (handing control to Science Observer) shall complete within 1 second of the request being approved. 3. Instrument configuration checks (Validate Program, Check Instrument Configuration) shall complete within 5 seconds .before execution begins. Time spent waiting for the Science Observer’s decision in UC-01 is excluded.
Basic Workflow:	1. The Telescope Operator selects a submitted observing program to execute. 2. The system validates the program (see 2a, 2b). 3. The system checks that the instrument is properly configured. 4. The system checks that the Weather Monitoring System indicates safe observing conditions. 5. The system checks that no other subsystem currently holds a lock/control conflict. 6. The Telescope Operator commands the system to begin observing, and the system starts executing the program in the selected mode (Planned, Queue, or Service). 7. During observation, the Science Observer may request to switch to Interactive Mode to issue commands directly
Alternative Workflow:	2a — Program fails validation If the system cannot validate the observing program (e.g., missing parameters, invalid target), the system rejects the start request and notifies the Telescope Operator with the reason. The use case ends. 2b — Soft error detected during validation Validate Program is extended by Confirm Soft-Error Warning (UC-01). If the Science Observer confirms, validation completes and

	the flow continues at step 3. If the Science Observer does not confirm (declines, times out, or loses connection), the system does not start this program, notifies the Telescope Operator, skips the program, and selects the next-best target. The use case ends for this program. 3a — Instrument not properly configured If the instrument configuration check fails, the system prompts the Telescope Operator to run configuration first (redirect to "Configure Planned Instrument Setup"), or automatically triggers Run Required Calibration if calibration is what's missing. 4a — Unsafe weather conditions If the Weather Monitoring System reports unsafe conditions, the system prevents the observation from starting and displays a warning. The Telescope Operator may wait and retry once conditions improve. 5a — Subsystem lock conflict If another subsystem currently holds a lock, the system denies the start request and notifies the Telescope Operator which subsystem is in conflict (may involve Take Control of Contested Subsystem as a separate use case). 6a — Start command fails (hardware/system fault) If the system encounters a fault while attempting to begin execution, Apply Fault Recovery Procedure is triggered, and the use case may end in a failed state or retry after recovery. 7a — Science Observer's mode-switch request is denied If the Telescope Operator denies the request to switch to Interactive Mode, execution continues in the original mode and the Science Observer is notified. 7b — Manual intervention required mid-observation If an issue arises during execution, the Telescope Operator may trigger Intervene in Running Observation to pause or adjust the observation without fully aborting. 7c — Observation must be aborted If a critical issue occurs (safety alarm, equipment fault, unsafe weather change), the Telescope Operator (or system automatically) triggers Abort Exposure, and the observation halts. Partial data may be preserved and the instrument returned to a safe state.

Category:	Observation Management
Risks:	1. The program fails to validate a. If the program fails to validate the observing program may not start 2. The instrument is not properly configured. a. leads to failure to start the observing program. 3. Unsafe weather conditions. a. the system prevent the observation from starting and displays a warning 4. The start command don't run properly a. leads to failure to start the observing program.
Possibilities:	1. Pre observation summary checklist a. Show all the pre-observation task as a checklist 2. Predictive weather buffer a. To show a predictive weather in the future. 3. Auto-resume after transient lock conflicts a. Instead of let the telescope operator manually retry the system if the subsystem lock make it auto


Login/Authenticated
Program Submitted
Validate Program
Validate Success
Validate Fail
Check Pre-Condition
No Pre-Condition Error
Pre-Condition Error
Start the Program
Check Interrupt
Got Interrupt
Resume
Denied
No Interrupt (completed)
Abort
Mode Switch Requested
Intervene
Aborted
Approved
Interactive Mode

4.Submit Observing Program

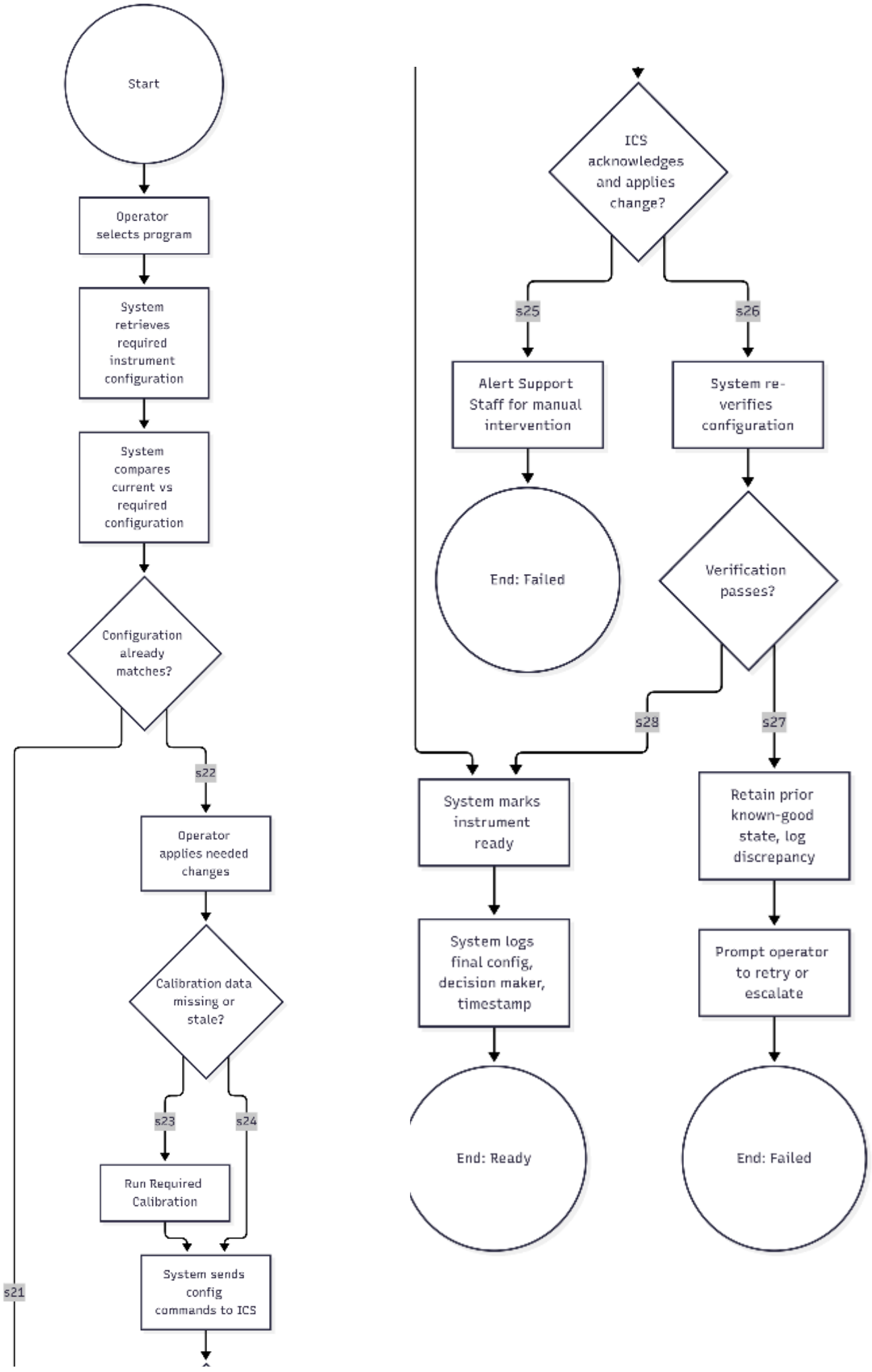
Use Case ID:	UC-04		
Use Case Name:	Submit Observing Program		
Process Owner:	Astronomer	Last Updated By:	Madhuri Singh
Date Created:	15/09/2026	Date Last Updated:	15/09/2026
Business Actor:	Astronomer		
Description:	Allows an Astronomer to submit an observing program that belongs to a science plan for further processing and scheduling.		
Trigger	The Astronomer has completed an observing program under a science plan and chooses to submit it.		
Relationship	Association: Astronomer		
Preconditions:	1. The Astronomer is logged in. 2. A science plan exists. 3. An observing program has been created under that science plan. 4. The observing program contains the required information.		
Postconditions:	The observing program is submitted, linked to its science plan, and stored in the system with status Submitted.		
Performance Goal:	The system shall submit the observing program and display the submission result within 2 seconds.		
Basic Workflow:	1. The Astronomer selects a science plan. 2. The Astronomer selects an observing program under that plan. 3. The Astronomer chooses Submit Observing Program. 4. The system checks the science plan and observing program information. 5. The system submits and records the observing program with its		

	science plan and sets its status to Submitted. 6. The system displays the submission result.
Alternative Workflow:	• 2a. Science Plan or Program Not Found: If the selected science plan does not exist, or the selected observing program does not belong to that science plan, the system displays an error message and asks the Astronomer to select a valid science plan and observing program. If none can be selected, the use case ends. • 4a. Incomplete Program Information: If the science plan or the observing program is missing required information, the system displays an error message and asks the Astronomer to complete the missing information before submission. • 4b. Invalid Program Information: If the science plan or the observing program contains invalid information, the system displays a validation message and asks the Astronomer to correct the information before submission. • 5a. Submission Failed: If the system cannot successfully submit the observing program due to a system or connection error, the system displays an error message, keeps the program status unchanged, and allows the Astronomer to retry the submission.
Category:	User-initiated
Risks:	Incomplete or invalid observing program information may prevent successful submission.
Possibilities:	The system could provide validation feedback before submission to help the Astronomer correct missing or invalid information.

5. Configure Planned Instrument Setup

Use Case ID:	UC-05		
Use Case Name:	Configure Planned Instrument Setup		
Process Owner:	Gemini Science Operations	Last Updated By	Nattanan Jumparuang
Date Created:	15/09/2026	Date Last Updated:	15/09/2026
Business Actor:	Telescope Operator		
Description:	Allows the Telescope Operator to configure the instrument physically (filter, grating, detector mode, etc.) prior to the execution of a program, so that the Start Observing Program configuration check is satisfied. Directly called or redirected to from UC-03 if check is not satisfied.		
Trigger:	The operator chooses a program and starts setup, or the system diverts to this point from a failed check in Start Observing Program.		
Relationship	Association : Telescope Operator, Instrument Control System (ICS), Support Staff		
Preconditions:	1. Operator login is successful. 2. The program with the required instrument configuration is selected. 3. The instrument is not locked with any exposures.		
Postconditions:	1. Configuration of the instrument conforms to the requirements of the program. 2. The configuration is recorded relative to the program. 3. The instrument is in the ready state (or fails and remains in the last good state).		
Performance Goal:	Configuration test done within 5 seconds.		
Basic Workflow:	1. The operator selects the program. 2. System fetches required configuration and checks against existing one. 3. The operator makes changes. 4. The system sends commands to the Instrument Control System. 5. ICS makes changes; system verifies again. 6. The system records results and instrument is marked ready.		

Alternative Workflow:	3a. Already matches → go to ready. 4a. Calibration missing/stale → activates Run Required Calibration. 5a. ICS not responding → notify Support Staff. 6a. Verification unsuccessful → hold previous status, record difference, re-verify.
Category:	Observation Management
Risks:	False “successful” report of hardware; concurrent operators at odds using the same machine; hurried verification due to time constraints.
Possibilities:	Daytime pre-stage/config verifications; re-usable config templates; closed-loop sensor verifications.


Start
Operator selects program
System retrieves required instrument configuration
System compares current vs required configuration
Configuration already matches?
s21
s22
Operator applies needed changes
Calibration data missing or stale?
s23
s24
Run Required Calibration
System sends config commands to ICS
ICS acknowledges and applies change?
s25
s26
Alert Support Staff for manual intervention
End: Failed
System re-verifies configuration
Verification passes?
s28
s27
System marks instrument ready
System logs final config, decision maker, timestamp
End: Ready
Retain prior known-good state, log discrepancy
Prompt operator to retry or escalate
End: Failed